

Radiation-Conditioned Ti-hBN Coatings for Space Mechanisms: Bridging Tribology and Irradiation Effects from Laboratory to Low-Earth Orbit

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Abstract

The longevity of tribological components in extraterrestrial environments is challenged by abrasive regolith, extreme temperatures, and ionizing radiation. This study evaluates the performance of vacuum plasma sprayed (VPS) Ti-2vol.% hexagonal boron nitride (hBN) coatings applied to joint mechanisms such as rod and slot, ball and socket, and hinge assemblies, fabricated from Al6061 and Ti6Al4V for planetary structures. Coated and uncoated samples were tested under vacuum with lunar regolith simulant (JSC-1A), followed by environmental exposures including electron radiation and thermal cycling. Results show that coated samples consistently outperformed uncoated counterparts across all configurations. Wear depth in coated components was reduced by over 80%, as confirmed by 3D profilometry and surface imaging. SEM analysis revealed severe abrasion and localized plastic deformation in uncoated surfaces, while coated samples retained structural integrity due to synergy of hard protective phases formed during deposition and retained hBN providing solid lubrication in the sliding surface. Coated and exposed samples exhibited smoother actuation profiles and reduced friction, indicating a beneficial "conditioning" effect from environmental exposure. The coatings-maintained performance even on complex geometries with minimal delamination. These findings demonstrate the potential of VPS Ti-hBN coatings to extend the operational life of lunar mechanisms, bridging laboratory-scale tribology with flight-relevant mechanical applications.

1.0 Introduction

The present work extends that coating development to functional mechanisms. VPS Ti-2 vol.% hBN was applied to three representative joint geometries - rod and slot, ball and socket, and hinge - and evaluated under vacuum actuation with JSC-1A regolith simulant. Coated, uncoated, exposed, and unexposed conditions were compared to determine whether the coating could be transferred from coupon-level tribology to realistic moving assemblies relevant to Artemis and Human Landing System architectures.

Ensuring the longevity and reliability of mechanical components in extraterrestrial environments is a critical challenge for planetary exploration missions. Structural components of space systems, including lunar and Martian rovers, landers, and surface habitats, are typically constructed from lightweight alloys such as Al 6061 and Ti6Al4V [1–5]. However, these materials often suffer premature failure due to extreme thermal fluctuations ranging from -196 to 150 °C, exposure to cosmic ionizing radiation such as Galactic Cosmic Rays (GCR) and Solar Particle Events (SPE), and secondary neutron radiation that leads to embrittlement and fracture[6–8]. Additionally, interactions with sharp and abrasive lunar and Martian regolith cause severe material degradation, further limiting the operational longevity of these components[9–11]. A significant concern in planetary exploration systems is the failure of critical mechanisms such as joints, hinges, gears, and actuators, which are essential for rover mobility, robotic arm articulation, and deployable structures. These tribological components experience substantial wear due to regolith ingress, leading to degradation and loss of function over time. Therefore, developing wear-resistant protective coatings for moving mechanical assemblies is vital for enhancing the durability and performance of planetary exploration systems.

Hexagonal boron nitride (hBN) has emerged as a promising material due to its exceptional tribological properties, including excellent thermal conductivity of 550 W/mK, a low friction coefficient ranging from 0.1

to 0.7, and outstanding wear resistance. Hexagonal boron nitride (hBN) is a promising reinforcement for space tribology because its layered crystal structure provides low shear and solid-lubricant behavior, while boron also contributes neutron attenuation capability. In earlier work, the authors compared APS and VPS Ti-hBN coatings at 2 and 10 vol.% hBN and identified VPS Ti-2 vol.% hBN as the best overall candidate because it provided the most favorable balance of hardness, plasticity, BN retention, radiation tolerance, and erosion resistance[12–14].

To address these challenges, previous studies developed and evaluated Ti/hBN composite coatings using both Atmospheric Plasma Spray (APS) and Vacuum Plasma Spray (VPS) deposition methods at two different compositions: 2 vol.% hBN and 10 vol.% hBN[18]. The coatings were systematically tested for their mechanical, tribological, and environmental resistance properties under simulated space conditions. From this series of studies, VPS Ti/2vol.% hBN was identified as the best candidate for a down-selected coating, as it demonstrated superior performance across key parameters of wear, radiation, and erosion.[3,9,10]. The 2 vol.% hBN composition provided the optimal balance between hardness and wear resistance, with enhanced resistance to abrasive wear, higher BN retention, and reduced formation of brittle oxide phases compared to higher BN concentrations. The VPS deposition method further improved coating density and adhesion, minimizing porosity and increasing mechanical integrity, while preserving sufficient plasticity to absorb impact forces during erosion. These attributes made VPS Ti/2vol.% hBN the most promising candidate for tribological applications in planetary mechanisms.

In previous work, VPS Ti/2vol.% hBN coatings demonstrated a significant improvement in abrasive wear resistance, achieving a 90% reduction in wear against lunar regolith JSC-1A, along with a 27% enhancement in neutron radiation shielding under controlled laboratory conditions[10]. Further testing under simulated lunar environmental conditions, including thermal cycling, electron radiation, and synergistic electron-thermal exposure, revealed up to a 90% reduction in wear and a 28% improvement in neutron shielding, attributed to enhanced neutron capture by the ^{10}B isotope[3]. Moreover, planetary erosion testing at extreme lunar temperatures from -196 to 150 °C confirmed that the coatings exhibited superior resistance to erosion compared to uncoated Ti6Al4V substrates, owing to their improved plasticity and high hardness[9].

Building upon these promising findings, the present study focuses on evaluating the tribological performance of VPS Ti/2vol.% hBN coatings applied to functional mechanisms such as hinges and joints that experience relative motion and contact under exposure to simulated radiation and thermal cycling. The primary objective is to develop and validate the VPS Ti/2vol.% hBN coating application process for mechanisms with complex geometries and assess their wear resistance under operationally relevant conditions. Following initial pin-on-disk wear tests to identify the optimal coating candidate, the selected coating was applied to more complex mechanical assemblies to determine its feasibility for tribological applications. A series of mechanisms were designed to evaluate the scale-up potential and robustness of the coating process while replicating different types of motion found in structural systems used during the Apollo program and planned future missions such as NASA's Human Lander System Program[19,20]. These mechanisms were configured to provide varying instances of wear at interacting surfaces, incorporating three distinct designs: rod and slot, ball and socket, and hinge.

The novelty of this study lies in the first-time application of VPS Ti/2vol.% hBN coatings on tribological mechanisms designed for planetary exploration, bridging the gap between laboratory-scale testing and real-world functional applications in extreme environments. By systematically investigating the tribological behavior of these coatings in moving mechanical assemblies, this study aims to establish their effectiveness in mitigating wear-related failures and advancing the durability of critical mechanisms for sustained lunar operations.

2.0 Materials and Methods

2.1 Mechanism Fabrication

A series of mechanical joint configurations were designed and fabricated to evaluate the scale-up potential and robustness of the VPS Ti-2%hBN coating process. These configurations were inspired by mechanisms historically used in the Apollo missions and those proposed for future lunar infrastructure such as NASA's Human Lander System (HLS) program[21–26]. The designs were intended to replicate distinct types of relative motion and wear mechanisms observed in operational lunar structures, allowing for tribological performance assessment under representative mechanical loads.

Three distinct joint configurations were developed: Rod and Slot, Ball and Socket, and Hinge. Each configuration was fabricated using two standard aerospace-grade materials: anodized aluminum alloy Al6061 (gold in appearance) and titanium alloy Ti6Al4V (silver in appearance). Both conventionally manufactured (CM) and additively manufactured (AM) parts were produced. AM specimens, made from a titanium-based powder, were included primarily to assess coating deposition compatibility on complex geometries, though they were not subjected to mechanical actuation tests due to size constraints and printing limitations. All specimens were designed with sufficient surface area to enable quantitative physical wear evaluation after environmental exposure and actuation.

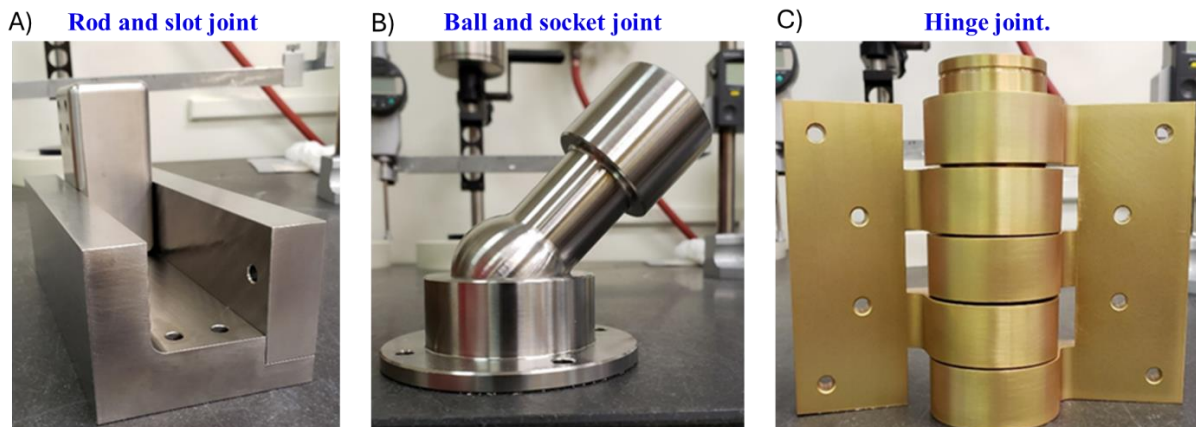


Figure 1. (A) Rod and slot joint simulating linear sliding contact, (B) Ball and socket joint enabling multi-axial articulation, and (C) Hinge joint replicating rotational pivot motion. These mechanism configurations were designed to evaluate the tribological performance of VPS Ti-2%hBN coatings under relative motion and simulated space environmental conditions. Components were fabricated using anodized Al6061 and Ti6Al4V to represent structural materials used in lunar mechanisms.

Rod and Slot Configuration: This configuration consisted of a flat, linear channel in which a rectangular rod could translate back and forth, simulating sliding contact (Figure 1A). For CM specimens, the rod's sliding face measured 2.00 inches in length by 1.75 inches in width, with a total slot travel distance of 6 inches in a single direction. For AM specimens, reduced dimensions were used to accommodate build platform constraints; the rod measured 1.50 × 1.25 inches and allowed for 2 inches of sliding travel.

Ball and Socket Configuration: In this configuration, a rod with a spherical ball end articulated within a hemispherical socket, allowing angular movement representative of gimbal-type joints (Figure 1B). Both the ball and socket components had a diameter of 1.75 inches, and the joint was designed to permit articulation up to 30 degrees from the central axis.

Hinge Configuration: The hinge joint consisted of two plates connected by a cylindrical pin to simulate pivoting motion (Figure 1C). CM hinge plates featured five ring segments with 1.75-inch diameter holes housing a 1.72-inch pin, allowing for 180 degrees of uniaxial rotation. In contrast, AM hinge plates were produced with three-ring segments, using 1.53-inch diameter holes and a corresponding 1.50-inch pin, scaled for additive manufacturing feasibility.

Each configuration was produced in both coated and uncoated versions to enable comparative tribological assessment. The coating applied was a vacuum plasma sprayed (VPS) Ti6Al4V matrix reinforced with 2 vol.% hexagonal boron nitrides (hBN), denoted as Ti-2%hBN-VPS. This composition was selected as the optimal candidate from a broader coating development campaign that included both APS and VPS deposition techniques and multiple BN loadings (2 and 10 vol.%), with VPS Ti-2%hBN demonstrating superior hardness, reduced oxide-induced brittleness, improved BN phase retention, and enhanced plasticity for erosion resistance. The detailed coating deposition parameters are provided in our previous publications on Titanium-Hbn coatings for lunar applications.

Initial coating trials on aluminum substrates exhibited melting and delamination due to thermal mismatch. To mitigate these issues, a nickel-based Ni-5Al interlayer was introduced between the aluminum substrate and the functional coating to improve thermal management and bonding integrity. This process adjustment

enabled consistent deposition and adhesion of the Ti-2%hBN coating across both planar and non-planar geometries. Mechanism sets were subsequently subjected to sequential environmental exposures, consisting of ionizing radiation followed by thermal cycling between -180 °F and +150 °F. These exposures were conducted independently to isolate their effects. Select specimens were left unexposed and/or uncoated to serve as experimental controls. This design enabled a detailed comparison of tribological behavior under conditions approximating those on the lunar surface, supporting the overall objective of validating the coating's functional performance in complex joint assemblies.

2.2 Test Setup and Design

Actuation testing of the coated and uncoated mechanisms was performed using a retrofitted vacuum plasma spray chamber (Plasma Process LLC, Huntsville, AL), shown in Figure 2A. Originally designed for coating components with similar geometries to the fabricated joints, the chamber was selected due to its compatibility with lunar regolith simulant and its ability to withstand abrasive particle exposure without incurring internal damage. The system was capable of achieving low-level vacuum pressures and featured an integrated actuation system with five degrees of freedom, allowing for diverse motion profiles representative of in-situ lunar mechanisms.

The chamber was mechanically adapted with modular connection interfaces to accommodate each of the three mechanism configurations—rod and slot, ball and socket, and hinge—as illustrated in Figure 2B-D. Instrumentation was incorporated to enable real-time data acquisition during the tests. This included an axial force transducer for the rod and slot configuration and both a force transducer and rotational torque sensor for the articulating configurations. The torque sensor constrained angular actuation to 180° to prevent entanglement of the data transmission cables, ensuring safe and uninterrupted measurement throughout the testing cycles.

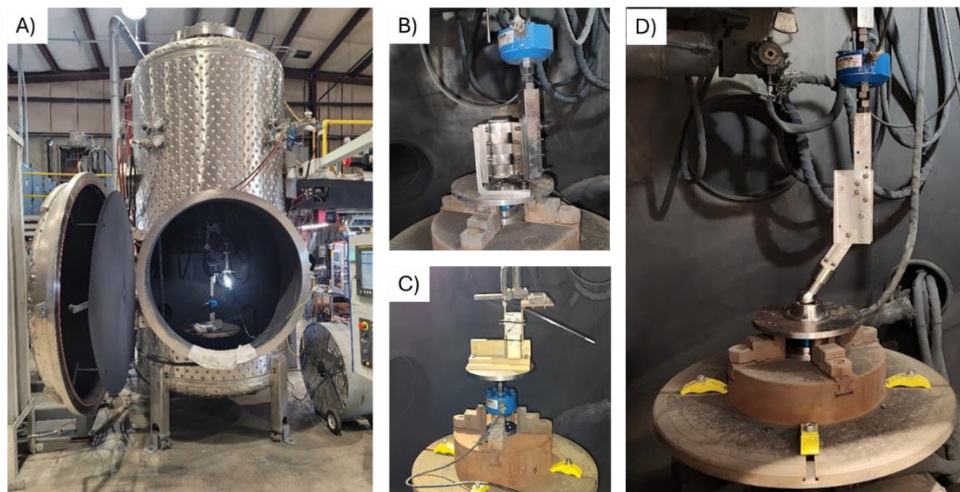


Figure 2: (A) Retrofitted vacuum plasma spray chamber used for actuation testing under low-pressure conditions with lunar regolith simulant. (B) Hinge joint mounted with axial force sensor inside the chamber. (C) Rod and slot joint setup integrated with axial load instrumentation. (D) Ball and socket joint configured with axial and torque transducers. Each mechanism was tested under 2–3 torr vacuum pressure, with regolith introduced during motion cycles to simulate tribological wear in a lunar environment.

Environmental conditions within the chamber were limited to pressure manipulation, with vacuum levels reaching approximately 2–3 torr during operation. While thermal control was not actively implemented due to the extensive chamber modifications, passive thermal variation was recorded, with internal temperatures fluctuating between 90 °F and 120 °F over the course of the test series. No active cooling or heating elements were employed. To visually monitor mechanism behavior and confirm proper operation throughout testing, a wireless camera was mounted inside or positioned externally to capture continuous video through a viewing port.

A pressure-fed powder feeder system, traditionally used for powder delivery during plasma spraying, was employed to introduce JSC-1A lunar regolith simulant into the interface of each joint. The system delivered the simulant at a controlled rate of approximately 1 g/min, simulating transient regolith ingestion that may occur during short-duration lunar operations. The simulant served as the primary abrasive agent to evaluate tribological degradation in each configuration. Prior to testing, each joint was subjected to a preload of 50–75 lbf axial force, followed by mechanical actuation over 500 full motion cycles, lasting approximately 2.5 hours under vacuum conditions. This protocol aimed to replicate operational wear mechanisms in lunar-like environments and provide insight into coating performance under combined mechanical and abrasive loading. To simulate harsh lunar conditions, the coatings were subjected to radiation, and thermal cycling, exposures at NASA Marshall Space Flight Center (MSFC) in Huntsville, AL. Radiation exposure was conducted using a Pelletron particle accelerator under high vacuum (1×10^{-6} Torr) with an electron beam energy of 400 keV, delivering a total dose of 6 Mrad. Thermal cycling was performed in the ET V3 thermal vacuum chamber, with samples exposed to temperature extremes ranging from -170 °C to +120 °C under vacuum. These conditions were designed to replicate or exceed the lunar environment, enabling accelerated evaluation of coating durability and performance. Detailed parameters for the exposure testing are provided in our previous publication[3].

2.3 Data Collection and Analysis

During testing, mechanical response data were recorded in real time. Axial load data were captured using an Interface Force Model 1000 load cell (capacity: 1000 lbf, accuracy: ± 1.000 lbf) mounted above the mechanism to monitor the vertical load imparted during actuation (Figure 2B-D). Rotational torque data were collected using an Interface Force TS11 Series torque sensor (rated for 177 lbf-in, accuracy: ± 0.177 lbf-in), mounted beneath the mechanism to record resistance during joint rotation. Post-test evaluation included mass loss measurements to assess material removal and coating degradation. Each specimen was weighed before and after testing to quantify total mass loss. High-resolution photographs were taken pre- and post-test to document visible wear patterns and any localized coating failure. In addition, surface topography of the wear tracks was characterized using a white light profilometer (Keyence VK-X1000) to quantify changes in surface roughness, coating thickness, and wear depth. A scanning electron microscope was employed (JEOL FSM -100) to investigate the microstructural features and mechanisms after wear.

This integrated testing setup enabled a comprehensive evaluation of coating performance under representative lunar operational conditions, combining mechanical loading, regolith abrasion, and vacuum exposure. However, the current testing approach presented several limitations that affect the precision of quantitative tribological analysis. Notably, the setup did not allow for reliable measurement of the coefficient of friction or wear rate due to the complexity of the joint geometries and the presence of lunar regolith simulant at the contact interfaces. The irregular distribution of regolith and resulting non-uniform wear further complicated direct force and frictional assessments. While these factors limited quantitative comparisons, qualitative evaluations using post-test imaging, profilometry, and SEM analysis still provided meaningful insights into coating performance under simulated lunar conditions.

3.0 Results and Discussions

To interpret the test data clearly, the following nomenclature (Table 1) was used for identifying samples:

Table 1: Nomenclature for identifying test samples in data collection

Identifier	Configuration
RS	Rod and Slot
BS	Ball and Socket
HG	Hinge
TI	Titanium alloy
C	Coated
UC	Uncoated
E	Exposed (Radiation + Thermal cycle)
UE	Unexposed
1, 2, 3, and so on.	Sample identifier

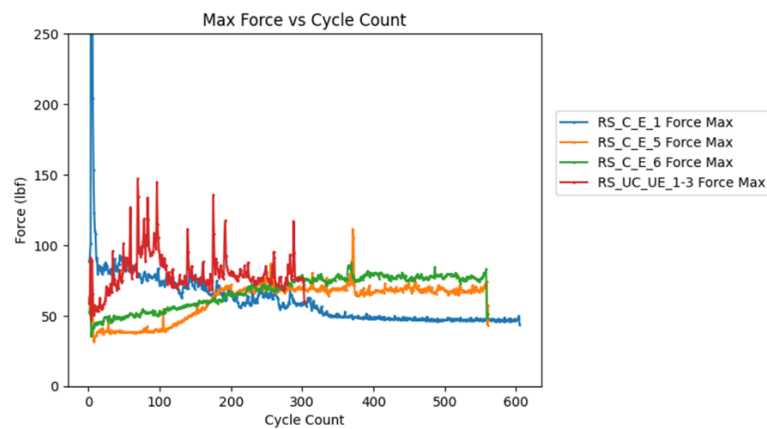
3.1 Rod and Slot Configuration

Figure 3A shows the maximum axial force per cycle for the Rod and Slot configuration. In coated and environmentally exposed samples such as RS_C_E_1, a progressive decline in peak force was observed.

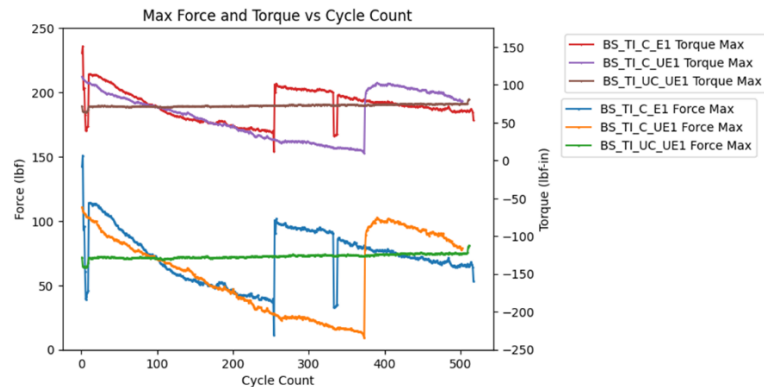
This trend suggests gradual wear and surface smoothing as the test progressed. Interestingly, the force required to push the rod forward was consistently higher than the force needed to retract it, indicating asymmetrical wear or friction possibly due to regolith accumulation. Despite some noise from sensor instability, the coated samples displayed smoother force trends, suggesting more controlled contact behavior.

In contrast, RS_C_E_5 and RS_C_E_6, which were also coated and exposed, showed increasing resistance over time. This likely resulted from mechanical interlocking between the hard coated rods and uncoated aluminum slots—where the softer substrate was more prone to regolith-induced surface distortion. Uncoated and unexposed samples RS_UC_UE_1–3 showed less predictable force trends, with waveform patterns that crested and dipped repeatedly. This can be attributed to evolving surface incongruities and abrasive wear from the regolith. These irregularities formed and were gradually worn down during cyclic motion, contributing to fluctuating force requirements. Additionally, severe abrasion in the uncoated samples generated noisy datasets, contrasting with the smoother profiles of coated specimens.

A) Maximum force per cycle for the rod and slot tests



B) Maximum axial force and torque for the ball and socket tests.



C) Maximum force per cycle for the Hinge tests

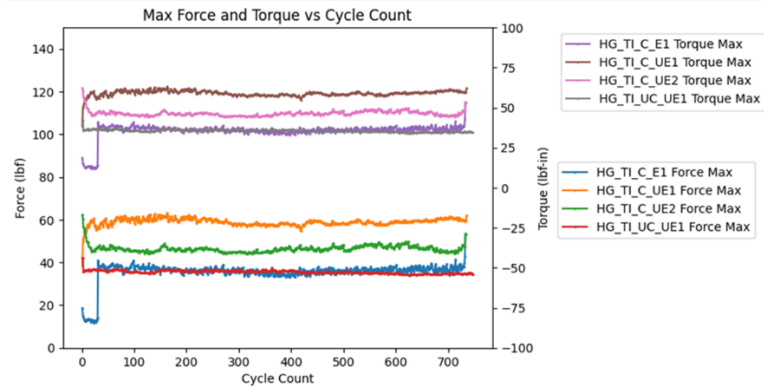


Figure 3: A) Maximum force per cycle for the rod and slot tests B) Maximum axial force and torque for the ball and socket tests. C) Maximum axial force and torque for the hinge tests.

3.2 Ball and Socket Configuration

Figure 3B presents axial force and torque data for the Ball and Socket configuration. The uncoated sample BS_TI_UC_UE1 maintained stable force and torque values throughout testing, as expected due to the absence of coating-induced friction or regolith interaction resistance. Coated samples BS_TI_C_E1 and BS_TI_C_UE1 began with higher torque and axial force values, resulting from tight tolerances and coating thickness in the mating surfaces. Once exposed to the regolith simulant, abrasive wear occurred, gradually reducing resistance. This was evident in the post-test physical inspection, where a distinct wear ring was observed on the coated ball surface—caused likely by socket edge intrusion and uneven loading.

Some tests required resets due to turntable overheating and thermal buildup from regolith insulation, visible as discontinuities in the data. These conditions underscore the tribological challenge of maintaining tolerances and wear resistance under combined mechanical and environmental stresses.

3.3 Hinge Configuration

The axial and torque trends for hinge mechanisms are shown in Figure 3C. The uncoated sample HG_TI_UC_UE1 demonstrated consistent actuation loads with minimal fluctuations across cycles. The coated and environmentally exposed sample HG_TI_C_E1 exhibited torque and force values comparable to the uncoated configuration, suggesting that the coating retained its structural and tribological performance even after radiation and thermal cycling. However, the unexposed coated samples HG_TI_C_UE1 and HG_TI_C_UE2 showed elevated torque and force values, likely due to the presence of an intact coating with sharper asperities not yet worn down by environmental or mechanical conditioning. This highlights the beneficial “wearing-in” effect of exposure, which may pre-condition the coating surface and reduce initial resistance.

3.4 Effect of Coating and Environmental Exposure Across All Conditions

Across all configurations, coated samples exhibited smoother trendlines, indicating more stable tribological behavior under actuation, especially when environmentally conditioned. In exposed samples, the coatings appeared to have adapted better through minor surface wear, smoothing asperities and reducing resistance. In contrast, uncoated samples, particularly in the Rod and Slot setup, experienced more severe wear, regolith-induced gouging, and higher signal noise. Coated but unexposed samples initially presented higher resistance, likely due to retained sharpness or incomplete contact conformity. However, post-exposure, the tribological performance improved as the coating surfaces adjusted under thermal and mechanical load. This behavior demonstrates that VPS Ti-2% hBN coatings, while initially stiff, develop more favorable surface conditions under operational conditions, suggesting strong potential for lunar application. A clear distinction was observed between the tribological behavior of coated samples that were exposed to radiation and thermal cycling and those that remained unexposed. Across all mechanism types, exposed coated specimens (e.g., RS_C_E_1, BS_TI_C_E1, HG_TI_C_E1) generally exhibited more stable and lower actuation forces and torques compared to their unexposed coated counterparts (e.g., RS_C_UE, BS_TI_C_UE1, HG_TI_C_UE2).

This behavior is attributed to the conditioning effect of environmental exposure: radiation and thermal cycling likely contributed to the relaxation of surface stresses, reduction of asperity sharpness, and partial smoothing of the coating surface—leading to reduced friction during articulation. In contrast, unexposed coated samples retained their as-sprayed surface roughness and tight clearances, which resulted in higher initial resistance, especially in sliding and rotational mechanisms. Moreover, synergistic exposure (combined radiation and thermal cycling) may have facilitated partial plastic deformation at the surface, enhancing conformity between mating components and further stabilizing contact interactions. These findings suggest that coatings benefit from pre-conditioning under environmental loads, and exposure may enhance tribological performance by accelerating surface adaptation prior to deployment in lunar environments.

3.5 Post-Test Inspection and Microstructural Analysis

Post-test visual inspection and surface analysis across all configurations—Rod and Slot, Ball and Socket, and Hinge—revealed consistent trends highlighting the performance benefits of the VPS Ti-2vol.% hBN coatings. Following each test, all mechanism components were visually inspected for damage, weighed, photographed, and analyzed using non-contact profilometry. Comparisons were made between coated and uncoated samples, as well as exposed versus unexposed specimens, across the rod and slot, ball and socket, and hinge configurations.

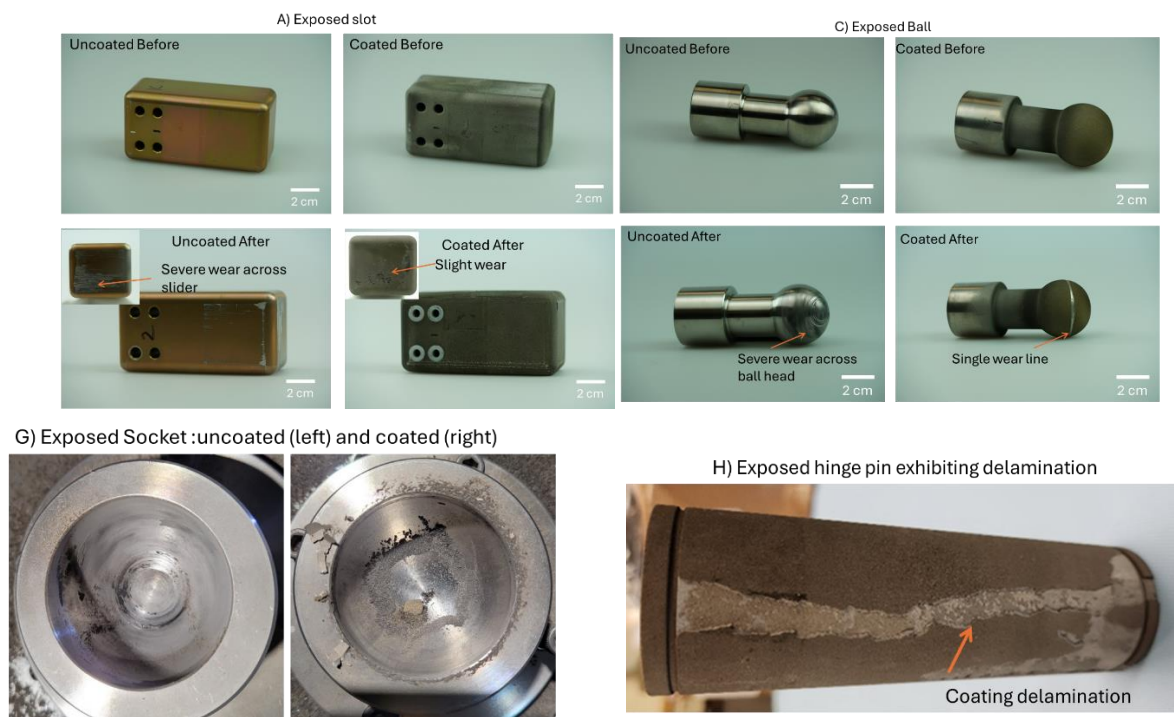


Figure 4: Representative post-test observations: slot wear comparison, ball wear comparison, socket condition, and an example of hinge-pin delamination associated with geometric/process complexity (A) Slot components showing severe abrasive wear on uncoated surfaces and only slight surface wear on coated counterparts, regardless of exposure. (C) Ball components with uncoated heads exhibiting broad wear scars, while coated heads developed a single, shallow wear line due to improved wear resistance. G) Ball and socket, socket post test, uncoated (left) and coated (right) showing regolith interaction H) Exposed hinge pin showing delamination

In the Rod and Slot setup, uncoated aluminum sliders displayed significant abrasive wear, particularly along the bottom surfaces, where regolith particles became embedded and contributed to accelerated material loss (Fig. 4A). In contrast, coated specimens retained much of their surface integrity, showing only minor polishing and localized wear. The wear was mostly superficial and did not penetrate the coating layer, validating the coating's ability to withstand regolith abrasion and mechanical sliding under lunar-relevant vacuum conditions. Importantly, no meaningful difference in wear behavior was observed between exposed

and unexposed coated samples, suggesting that environmental exposure (thermal cycling and radiation) did not degrade coating performance but may have even stabilized it.

The Ball and Socket configuration presented the most pronounced differences between coated and uncoated components. Uncoated ball heads exhibited severe, multi-latitude wear scars and material removal due to combined rotational loading and abrasive contact with regolith (Fig. 4C-D). In contrast, coated balls showed only a single, well-defined wear line, reflecting localized stress but substantially reduced material degradation (Fig. 4C). However, some edge chipping was noted in the coated balls, particularly near the transition from spherical to cylindrical geometry. This was primarily due to dimensional changes from coating thickness, which altered the intended contact geometry. The increase in ball diameter caused coated components to contact the socket edge rather than the spherical base, leading to concentrated wear and stress accumulation (Fig. 4G). Despite this, the coated sockets remained structurally intact, exhibiting only mild polishing and no spallation, in stark contrast to the uncoated sockets, which displayed extensive pitting, embedded regolith, and surface delamination. These findings underscore the coating's superior wear resistance but also highlight the need for design adjustments, such as adopting a conical socket geometry, to better accommodate dimensional shifts caused by coating application.

The Hinge configuration showed minimal wear overall due to its simpler uniaxial motion and reduced test load (limited to 40–60 lbf due to motor constraints). Still, clear distinctions were observed between uncoated and coated specimens. Uncoated hinge pins exhibited distinct linear wear lines, particularly in exposed samples (Fig. E-F), while coated pins showed only faint or slight wear marks, indicating better preservation of surface topography and tribological performance. The results confirm the effectiveness of the coating in resisting pivot-related degradation, especially in mechanisms designed for repeated actuation in abrasive environments.

The Figure 5 illustrates the dramatic difference in wear resistance between uncoated and VPS Ti-2vol.% hBN-coated Aluminum 6061 samples subjected to abrasive wear from JSC-1A lunar regolith simulant. The samples were taken from the "slot" components of the rod and slot mechanism following the completion of tribological testing in vacuum under actuation. The top row shows photographic images of the worn surfaces. The uncoated sample (left) exhibits large, visible wear scars and severe material loss along the sliding path. In contrast, the coated sample (right) retains a relatively intact surface with localized, shallow pitting and far less visual evidence of regolith-induced damage. A 10 × 10 mm region of maximum wear was selected on each surface (highlighted in yellow) for non-contact profilometry. The middle row presents 3D profilometry maps of the selected regions. In the uncoated part, a deep, continuous wear groove is visible along the sliding path, indicative of aggressive third-body abrasion from regolith particles. The coated surface, however, maintains most of its original topography, with no continuous wear track and only minor disruptions in the coating surface.

It is also important to note that a fair quantitative comparison of wear severity across different mechanisms is improbable in this study compared to conventional wear studies, as wear behavior is inherently dynamic, random, and location-specific for these complex mechanisms. Additionally, due to the complex geometries of the joint configurations, it is not possible to accurately measure wear volume for direct comparison, as factors such as local contact area, regolith impingement angle, and motion constraints vary significantly between designs. However, to demonstrate the effectiveness of the VPS coatings, wear depth profiles were obtained from the rod and slot configuration selected for its simple linear geometry and clearly defined maximum wear zone to enable a representative quantitative comparison of coated versus uncoated surfaces. Quantitative wear depth profiles along the red transect lines are shown in the bottom row. The uncoated sample exhibits a maximum wear depth of 160 μm, consistent with substantial material removal under repeated mechanical loading. The coated sample shows a significantly reduced wear depth of only 30 μm, confirming the enhanced surface durability and protective capability of the Ti-hBN coating. These results strongly support the efficacy of VPS-applied Ti-2vol.% hBN coatings in mitigating wear under lunar analog conditions, reducing surface damage by over 80% compared to uncoated aluminum. This performance is attributed to the coating's increased hardness, solid lubrication from retained hBN, and improved plasticity under load—all essential for extending the service life of tribological components in space environments[3,9,10]. Post-test SEM observations of the Al 6061 slot surfaces provided direct microstructural evidence of the stark contrast in wear behavior between uncoated and VPS Ti-2vol.% hBN-coated samples. In the uncoated condition, surfaces exhibited widespread mechanical degradation. As shown in Figure 6A–C, the worn region was dominated by severe abrasive scars, with deep furrows and intense plastic deformation aligned along the sliding direction.

Aluminum 6061 coated w/ 2%hBN-Ti VPS, 2 in. x 2 in. x 1 in. rectangular block “slot” mechanism part

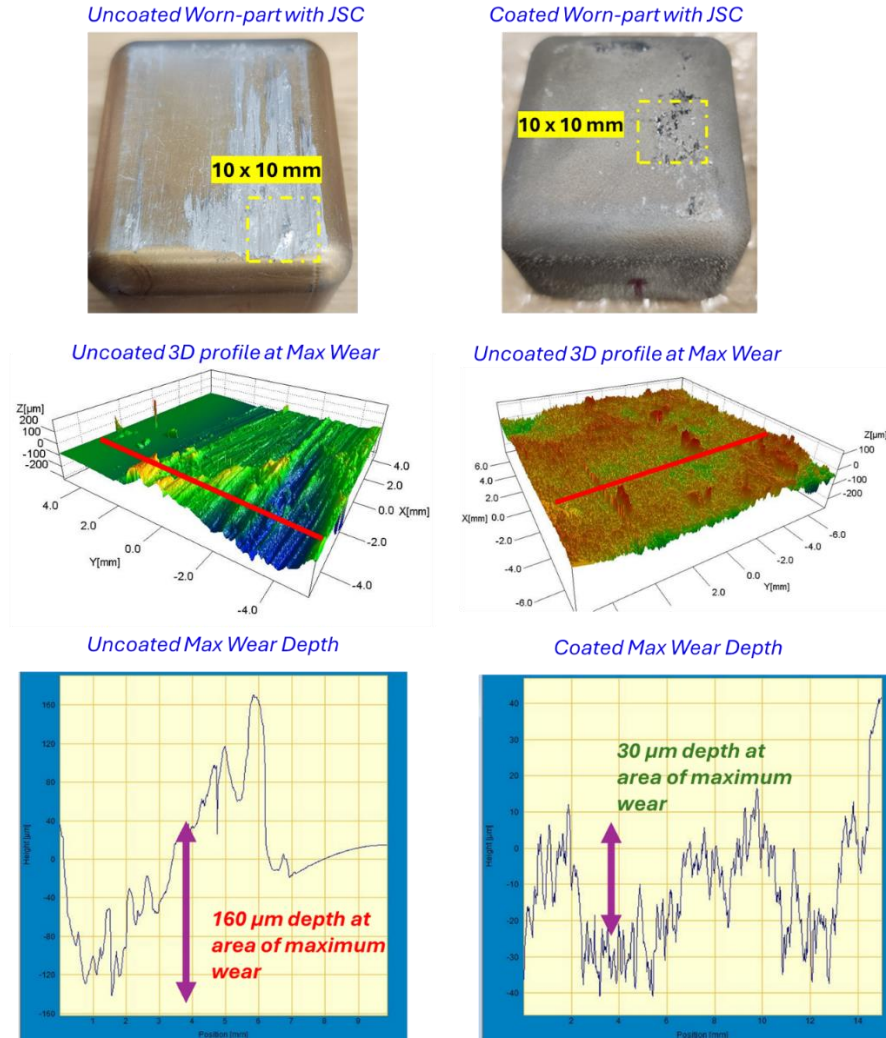


Figure 5: Comparison of wear behavior in uncoated and VPS Ti-2vol.% hBN-coated Aluminum 6061 "slot" mechanism components after exposure to JSC-1A lunar regolith simulant. (Top row) Optical images of worn surfaces showing a 10 × 10 mm area selected for profilometry analysis. The uncoated part shows extensive gouging and material removal, while the coated part retains surface integrity with minor pitting. (Middle row) 3D surface profilometry maps of the selected wear regions reveal severe trough formation in the uncoated sample and minimal surface disruption in the coated sample. (Bottom row) Wear depth profiles along the red lines in the 3D maps show a maximum wear depth of 160 μm in the uncoated sample versus only 30 μm in the coated sample.

High-magnification imaging (Figure 6B) revealed embedded JSC-1A regolith particles actively plowing through the aluminum substrate, acting as aggressive third-body abrasives[11]. The surface lacked any protective tribolayer, and in some areas (Figure 6C), oxidative wear was observed, likely initiated by the heat generated from high friction under vacuum and simulant exposure. These features collectively highlight the susceptibility of bare aluminum to rapid and severe wear under lunar-relevant tribological loading.

Conversely, the coated specimens displayed markedly improved surface retention and lower wear severity. SEM micrographs (Figure 6D–F) showed largely unworn coating splats (Figure 6D), interspersed with shallow grooves and minimal signs of plastic flow (Figure 6E). The enhanced wear resistance stems from the microstructural synergy between hard phases—such as TiO₂, TiN, and TiB₂—and embedded hBN flakes[3,18,27,28]. The hard phases reinforce the coating against abrasive intrusion, while the hBN (Figure

6F) contributes to a self-lubricating tribolayer that reduces interfacial shear stress and inhibits third-body particle embedding. These two mechanisms act cooperatively: the hard ceramic network bears mechanical load, while the hBN maintains a lubricating film that mitigates adhesive and abrasive interactions at the contact interface.

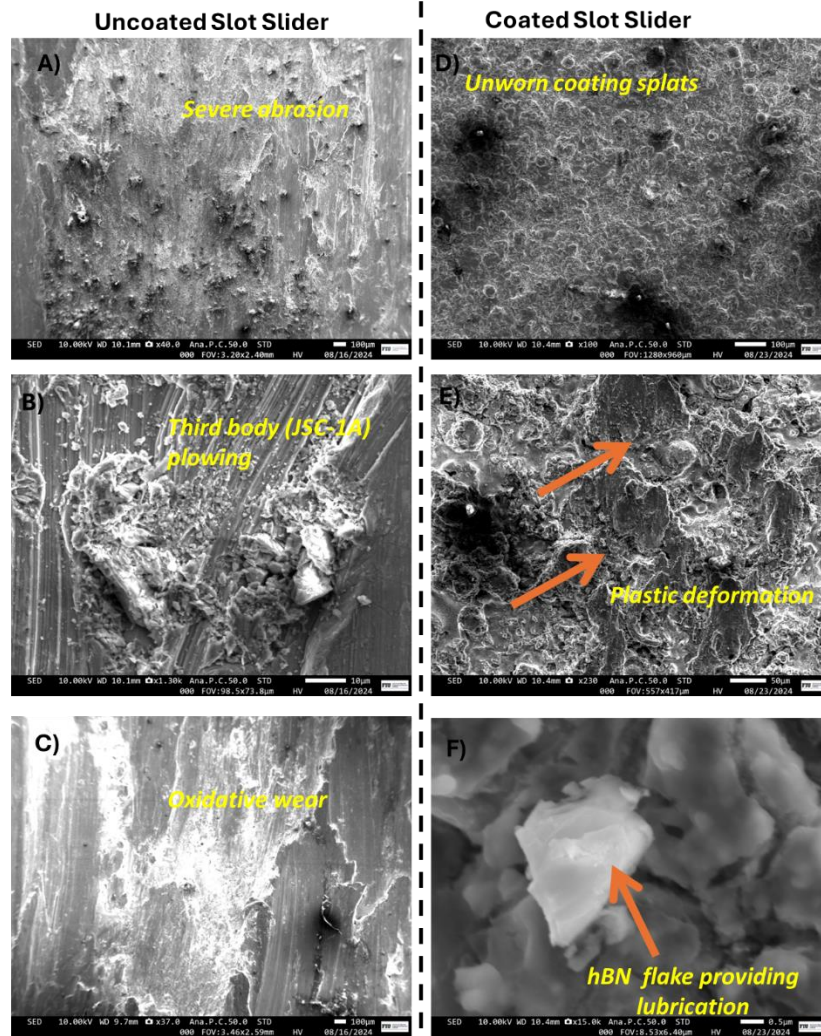


Figure 6: SEM Images of Uncoated and Coated Al 6061 Slot Surfaces After Regolith Wear Testing. (A–C) Uncoated samples show (A) severe abrasion, (B) third-body plowing by JSC-1A, and (C) oxidative wear. (D–F) Coated surfaces exhibit (D) unworn splats, (E) minor plastic deformation, and (F) hBN flakes aiding lubrication. The combination of hard phases and retained hBN significantly improved wear resistance and reduced regolith embedding.

Furthermore, the presence of unworn regions in coated samples underscores the localized nature of regolith damage and the coating's ability to deflect wear energy across the splat-matrix structure. Unlike the uniform material loss seen in uncoated aluminum, the coated surfaces retained integrity over much of the tested area, validating the coating's ability to delay the onset of failure even under aggressive sliding and abrasive conditions. These trends were consistently observed across all three mechanism configurations, rod and slot, ball and socket, and hinge, confirming the coating's effectiveness regardless of motion type or geometry. Importantly, samples exposed to radiation and thermal cycling retained their wear resistance and structural integrity, showing no signs of degradation due to environmental exposure. As established in our previous studies, radiation exposure introduces dislocation structures within the metallic matrix that can enhance mechanical strength and hardness, which likely contributed to the

sustained or improved wear resistance observed in the exposed coated samples compared to their unexposed counterparts[3,18].

A particularly notable outcome of this study is the favorable performance of coated samples following exposure to radiation and thermal cycling. Across all configurations, coated and exposed specimens outperformed their uncoated and even unexposed counterparts, revealing that environmental stress did not degrade the coating's properties. On the contrary, synergistic exposure likely promoted beneficial effects such as stress relaxation, partial plasticization, or smoothing of asperities thereby enhancing contact conformity and reducing friction[3,29]. These findings are encouraging for lunar applications, where components must endure long-term exposure to harsh extraterrestrial conditions. Conversely, uncoated and unexposed samples suffered from rapid surface degradation and regolith embedding, showing that even brief exposure to lunar simulant and actuation is sufficient to initiate failure in unprotected surfaces. This validates the need for robust protective coatings in lunar mechanisms to ensure long-term operational integrity.

3.6 Lessons Learned and Points for Improvement

Coating Process Adaptation to Complex Geometries: Translating the VPS Ti-2vol.% hBN coating process from planar samples to complex mechanism geometries presented several key challenges. The disparity in coefficients of thermal expansion (CTE) between titanium coatings and aluminum substrates resulted in coating delamination, substrate melting, and non-uniform thickness distribution, especially near sharp edges and corners. These failures, illustrated in early post-test examinations (Fig. 4H), were addressed through iterative refinements in the thermal environment, adjustments to spray angle and feed rate, and the introduction of a nickel-based interlayer. The interlayer served to buffer thermal mismatch and significantly improved coating adhesion and surface integrity.

Despite these advancements, the as-sprayed coatings had high surface roughness, leading to elevated friction during fit checks and actuation. Roughness was reduced to ~125 Ra through light sanding; however, even with surface finishing, coated joints exhibited higher actuation forces compared to uncoated ones. This suggests that additional improvements in coating finishing—such as polishing, lapping, or peening—are necessary to reduce frictional resistance and optimize tribological behavior for flight-ready components.

Infrastructure and Test Design Limitations: The testing infrastructure also imposed several limitations. The vacuum chamber, initially intended to operate at $\sim 10^{-3}$ Torr, was constrained to 2–3 Torr due to the regolith injection system. This deviation likely impacted regolith suspension and deposition, influencing the tribological environment. In the Rod and Slot configuration, the long vertical offset between actuator and joint interface introduced significant bending, reducing system stiffness and producing chatter during sliding motion. These dynamic artifacts likely skewed force measurements and led to nonuniform wear patterns. Removing one of the slot walls helped alleviate over constraint but did not fully eliminate structural compliance.

Further complications arose from actuator reliability. The linear stage motor failed during testing and had to be replaced, while the rotary motor experienced thermal overload, exacerbated by heat buildup and regolith insulation. To preserve equipment, axial loads had to be reduced mid-testing, likely limiting the full extent of wear in some components and reducing test repeatability. Also, A fair quantitative comparison of wear across mechanisms is limited due to dynamic, location-specific wear behavior and the complex geometries of each design, which prevent accurate wear volume measurement and introduce variability in contact area, motion, and regolith interaction. These issues highlight the need for robust, high-torque actuation systems, rigid fixturing, and the development of appropriate methods to accurately measure frictional forces, coefficient of friction, and wear volume in mechanisms with complex and non-uniform geometries.

4.0 Conclusions

Vacuum plasma sprayed (VPS) Ti-2 vol.% hBN coatings significantly enhance the wear resistance and environmental durability of tribological components used in lunar mechanisms. Across rod-slot, ball-socket, and hinge configurations, coated samples demonstrated superior surface preservation, improved actuation stability, and over 80% reduction in wear depth compared to uncoated components under abrasion from JSC-1A lunar regolith simulant. Force and torque measurements showed smoother and more stable trends in coated and environmentally exposed samples, indicating reduced friction and improved contact behavior. In contrast, uncoated specimens exhibited irregular force responses, severe abrasion, and regolith-induced gouging. Microstructural analysis revealed that hard ceramic phases such as TiO, TiN, and TiB₂ reinforce

the coating against abrasive penetration, while retained hBN forms a lubricating tribolayer, reducing shear stresses and limiting regolith particle embedding.

Despite these advantages, several practical considerations emerged during mechanism testing. The relatively high roughness of the as-sprayed coating increased initial actuation loads in unexposed samples, indicating the need for improved post-deposition finishing processes such as polishing or lapping. In addition, coating thickness altered component tolerances in some joints, highlighting the importance of incorporating geometric compensation during mechanism design. Test infrastructure limitations, including actuator alignment, motor heating, and vacuum constraints, also influenced load measurements. Overall, the results demonstrate that VPS Ti-2 vol.% hBN coatings provide a robust and scalable solution for protecting lunar mechanisms, significantly reducing abrasive wear while maintaining structural integrity under radiation, thermal cycling, and regolith exposure.

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